



**PROPEL-1K Program
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Alkali Hydroxide Triple Phase Flow Batteries

PI Matthew McDowell, Georgia Tech
Co-PI Asegun Henry, MIT

Project Vision

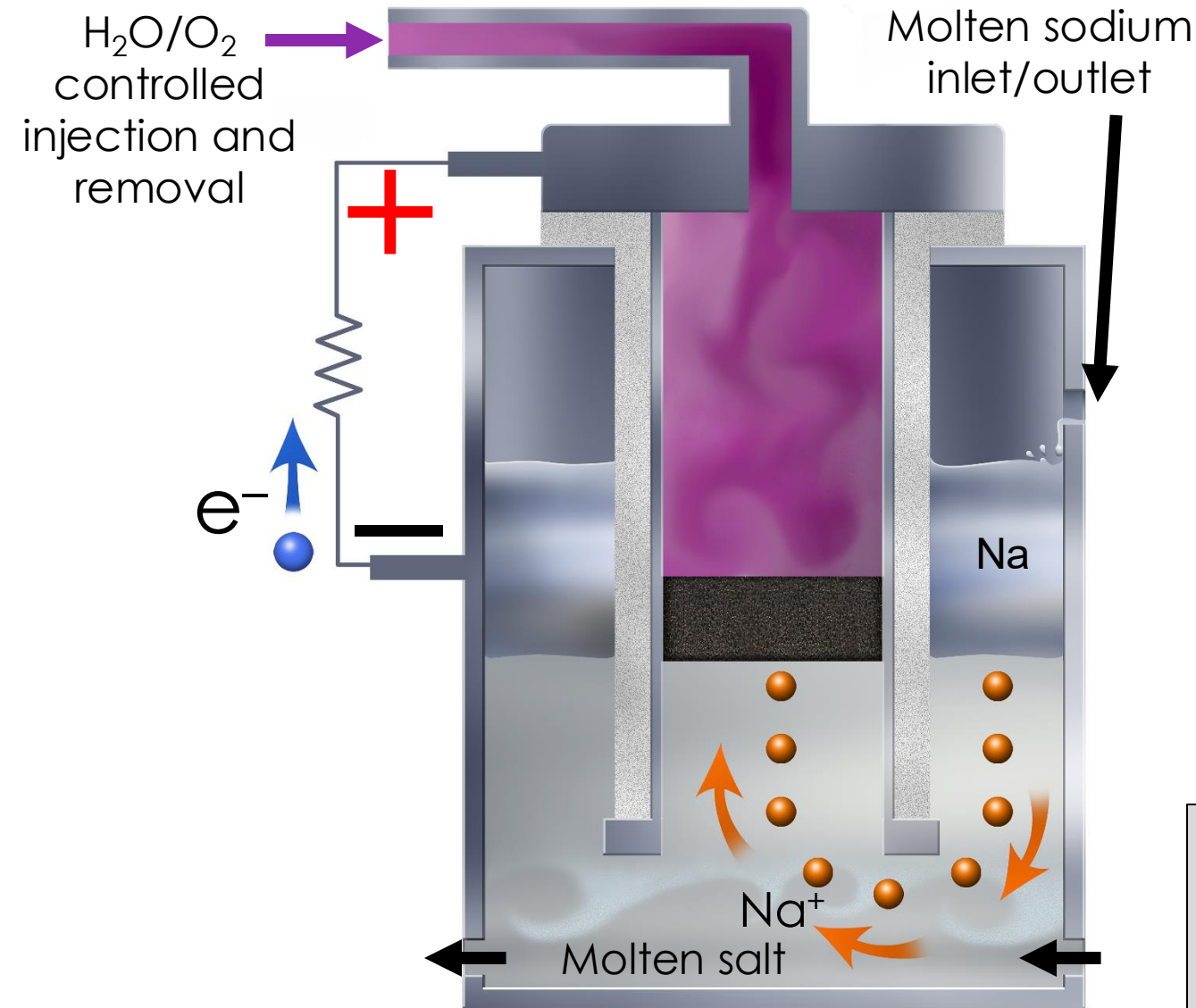
Enabling efficient, low-cost rechargeable alkali metal batteries via novel handling of molten metals and salts



**Massachusetts
Institute of
Technology**



Technology Development



1.55 kWh/kg (theoretical)

>1 kWh/kg achieved, considering active materials, separator, and current collectors.

Value proposition:

- Electrically or mechanically rechargeable
- Many individual cells can be used to provide power with high depth of discharge using flowing active materials

Technical achievements:

- Designed/constructed cells and demonstrating charge-discharge cycling
- Achieved >1 kWh/kg

Project success will enable rechargeable battery system with flexible energy and power capabilities for aircraft, ships, ESS.

Project Progress and Next Steps

Progress:

- Cell design and material handling
- Electrochemical reaction control
- Material stability at elevated temperature
- Reversible charge/discharge cycling

Remaining steps:

- Long-term testing and ALT
- Scaled system engineering
- Optimizing energy/power density
- Systems integration – aviation or maritime



Team and Potential Partnerships



Expertise: Electrochemistry, battery characterization, material stability analysis, battery engineering



Expertise: High-temperature chemistry and fluid pumping, thermal transport, scaled system engineering

Potential Partnerships:

- Long-term testing, accelerated life testing, and safety quantification
- Systems integration – aviation or maritime
- Scaled system engineering
- Demonstration in applications



Prof. Matt McDowell
PI, Georgia Tech



Prof. Asegun Henry
Co-PI, MIT



Dr. Mehdi Pishahang



Dr. Ahmad Harzandi



Dr. Sun Geun Yoon



Dr. Isuri Dammulla



Dr. Mouhamad Diallo

Technology-to-Market Approach

Value Proposition:

- ▶ We will achieve high energy density by enabling virtually all reactants to be converted via an intermediate-temperature fluid pumping system.
- ▶ We expect low cost per kWh based on ultra-low raw material costs and well-established manufacturing processes.
- ▶ Electrical charge and discharge with existing infrastructure.

Anticipated first users: EVTOL or other aircraft/drone

Potential long-term users: aviation, shipping, stationary storage